ENV 872L Final Project

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INSTRUCTIONS

```
#1 text each other before and after coding from our ends bc two of us can't work simultaneously
#2 pull before working
# push after working
# make the commit comments as detailed as possible bc grading will depend on that
```

What to focus on/ DELETE FOR FINAL OUTPUT

```
# Limiting data from January 1st, 2003 to December 31st, 2023 because we don't have full year's data fo
# Filter for columns (6): Location, Date/Time (in ymd format), O3, PM 2.5, PM 10, AQI
# Column names: Location, Date, O3, PM2.5, PM10, AQI
# Might need extra columns for Month and Year separately BUT AFTER INITIAL WRANGLING
```

INITIAL DATA WRANGLING

```
library(dplyr); library(lubridate); library(tidyverse);
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats 1.0.0     v stringr 1.5.1
## v ggplot2 3.5.1     v tibble  3.2.1
## v purrr   1.0.2     v tidyr   1.3.1
## v readr   2.1.5
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here); library(zoo); library(trend);
```

```
## here() starts at /home/guest/872L/ShuklaRahman
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(Kendall); library(tseries); library(ggplot2)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
here()
```

```
## [1] "/home/guest/872L/ShuklaRahman"
```

```r
Delhi_dataset_raw <- read.csv(here(
  "DelhiDataset/delhi_air_quality_feature_store_processed.csv"),
  stringsAsFactors = TRUE)

Delhi_dataset_summarized <- Delhi_dataset_raw %>%
  select(city, event_timestamp, pm25, pm10, o3, aqi) %>%
  mutate(event_timestamp = ymd_hms(event_timestamp)) %>%
  filter(event_timestamp >= ymd("2003-01-01") &
    event_timestamp <= ymd("2023-12-31")) %>%
  mutate(month = month(event_timestamp),
         year = year(event_timestamp)) %>%
  group_by(month, year) %>%
  summarize(
    PM2.5 = mean(pm25, na.rm = TRUE),
    PM10 = mean(pm10, na.rm = TRUE),
    O3 = mean(o3, na.rm = TRUE),
    AQI = mean(aqi, na.rm = TRUE)
  ) %>%
  mutate(Location = "Delhi")

write.csv(Delhi_dataset_summarized,
          here("processed/delhi_final_processed.csv"), row.names = FALSE)
```

```r
#reading csv file
BD_dataset_raw <- read.csv(here("AQI Bangladesh.csv"),
        stringsAsFactors = TRUE)

#filtering for Dhaka only
dhaka_dataset_raw <- BD_dataset_raw %>%
  filter(city_name == "Dhaka")

view(dhaka_dataset_raw)

#filtered for date,location,pm2.5,pm10,aqi,o3
dhaka_air <- dhaka_dataset_raw %>%
  mutate(datetime = ymd_hm(datetime)) %>%
  filter(city_name == "Dhaka") %>%
  select(city_name, datetime, ozone, pm2_5, pm10, aqi) %>%
  rename(
    Location  = city_name,
    Date = datetime,
    O3        = ozone,
    PM2.5     = pm2_5,
    PM10      = pm10,
    AQI       = aqi
  )

# Filter date range
dhaka_air <- dhaka_air %>%
  filter(Date >= as.Date("2003-01-01") & Date <= as.Date("2023-12-31"))

dhaka_dataset_summarized <- dhaka_air %>%
  mutate(month = month(Date),
         year  = year(Date)) %>%
  group_by(month, year) %>%
  summarize(
    PM2.5 = mean(PM2.5, na.rm = TRUE),
    PM10  = mean(PM10,  na.rm = TRUE),
    O3    = mean(O3,    na.rm = TRUE),
    AQI   = mean(AQI,   na.rm = TRUE)
  ) %>%
  mutate(Location = "Dhaka")

view(dhaka_dataset_summarized)

write.csv(dhaka_dataset_summarized,
          here("processed/dhaka_final_processed.csv"), row.names = FALSE)
```

```r
#reading both datasets
dhaka_final_processed <- read.csv(here("processed/dhaka_final_processed.csv"))
delhi_final_processed <- read.csv(here("processed/delhi_final_processed.csv"))

#combining into one dataframe
combined_dataset <- rbind(dhaka_final_processed, delhi_final_processed)

combined_dataset <- mutate(combined_dataset,
```

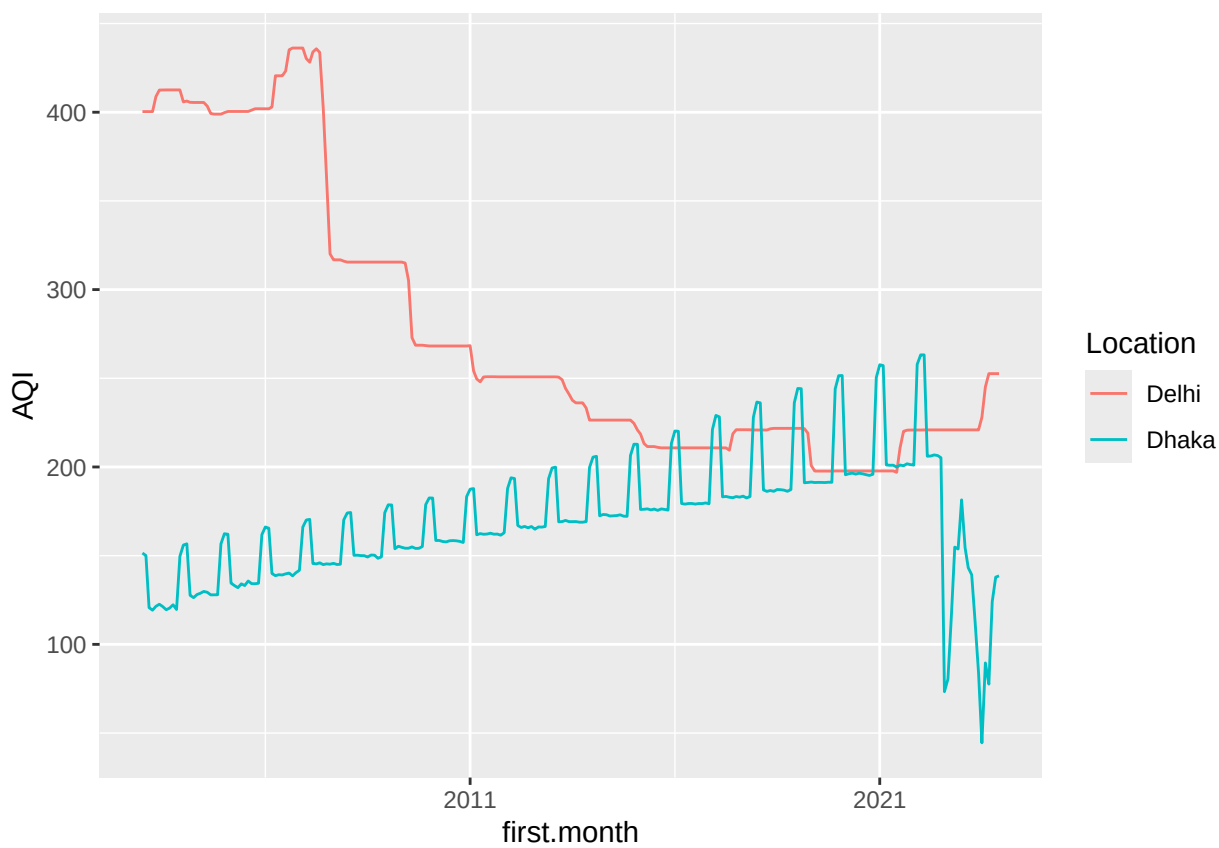
```
                                first.month = make_date(year, month, 1))

#checking for missing data
colSums(is.na(combined_dataset))
```

```
##       month        year       PM2.5        PM10          O3         AQI
##           0           0           0           0           0           0
##    Location first.month
##           0           0
```

TIME SERIES ANALYSES

```
#AQI
ggplot(combined_dataset, aes(y = AQI, x = first.month, colour = Location)) +
  geom_line() +
  scale_x_date(date_breaks = "10 years", date_labels = "%Y")
```



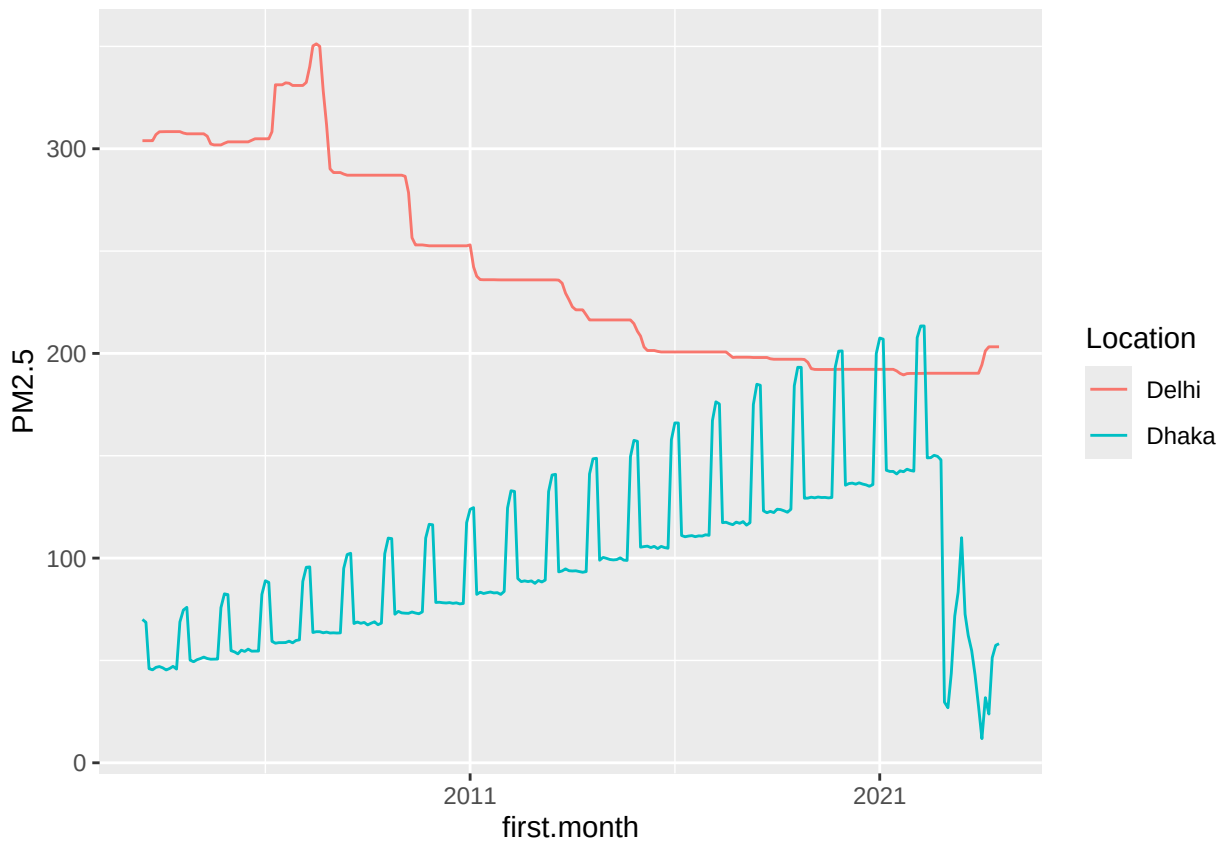

```
  geom_smooth(method = lm)
```

```
## geom_smooth: na.rm = FALSE, orientation = NA, se = TRUE
## stat_smooth: na.rm = FALSE, orientation = NA, se = TRUE, method = function (formula, data, subset, we
## {
```

```
##     ret.x <- x
##     ret.y <- y
##     cl <- match.call()
##     mf <- match.call(expand.dots = FALSE)
##     m <- match(c("formula", "data", "subset", "weights", "na.action", "offset"), names(mf), 0)
##     mf <- mf[c(1, m)]
##     mf$drop.unused.levels <- TRUE
##     mf[[1]] <- quote(stats::model.frame)
##     mf <- eval(mf, parent.frame())
##     if (method == "model.frame")
##         return(mf)
##     else if (method != "qr")
##         warning(gettextf("method = '%s' is not supported. Using 'qr'", method), domain = NA)
##     mt <- attr(mf, "terms")
##     y <- model.response(mf, "numeric")
##     w <- as.vector(model.weights(mf))
##     if (!is.null(w) && !is.numeric(w))
##         stop("'weights' must be a numeric vector")
##     offset <- model.offset(mf)
##     mlm <- is.matrix(y)
##     ny <- if (mlm)
##         nrow(y)
##     else length(y)
##     if (!is.null(offset)) {
##         if (!mlm)
##             offset <- as.vector(offset)
##         if (NROW(offset) != ny)
##             stop(gettextf("number of offsets is %d, should equal %d (number of observations)", NROW(o
##     }
##     if (is.empty.model(mt)) {
##         x <- NULL
##         z <- list(coefficients = if (mlm) matrix(NA, 0, ncol(y)) else numeric(), residuals = y, fitte
##         if (!is.null(offset)) {
##             z$fitted.values <- offset
##             z$residuals <- y - offset
##         }
##     }
##     else {
##         x <- model.matrix(mt, mf, contrasts)
##         z <- if (is.null(w))
##             lm.fit(x, y, offset = offset, singular.ok = singular.ok, ...)
##         else lm.wfit(x, y, w, offset = offset, singular.ok = singular.ok, ...)
##     }
##     class(z) <- c(if (mlm) "mlm", "lm")
##     z$na.action <- attr(mf, "na.action")
##     z$offset <- offset
##     z$contrasts <- attr(x, "contrasts")
##     z$xlevels <- .getXlevels(mt, mf)
##     z$call <- cl
##     z$terms <- mt
##     if (model)
##         z$model <- mf
##     if (ret.x)
##         z$x <- x
```

```
##     if (ret.y)
##         z$y <- y
##     if (!qr)
##         z$qr <- NULL
##     z
## }
## position_identity
```

```
#PM2.5
  ggplot(combined_dataset, aes(y = PM2.5, x = first.month, colour = Location)) +
  geom_line() +
  scale_x_date(date_breaks = "10 years", date_labels = "%Y")
```

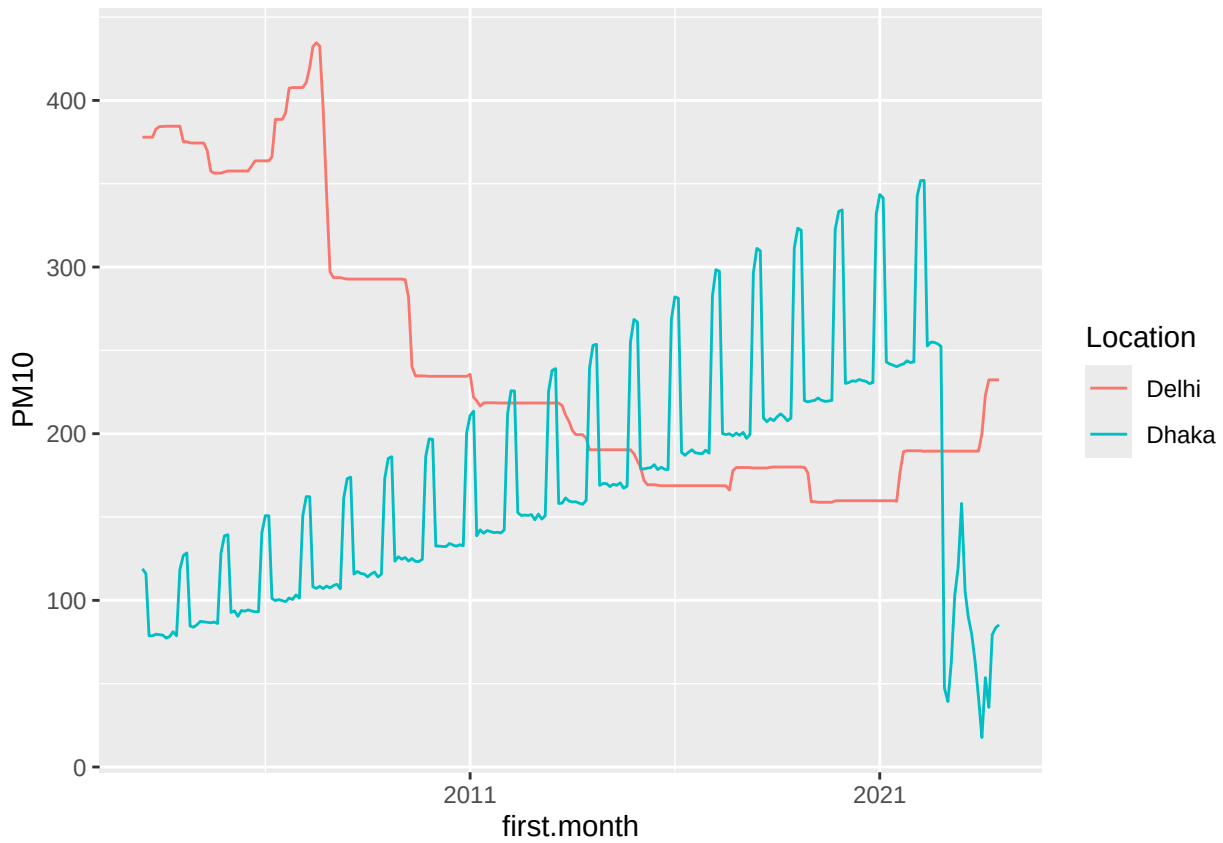



```
  geom_smooth(method = lm)
```

```
## geom_smooth: na.rm = FALSE, orientation = NA, se = TRUE
## stat_smooth: na.rm = FALSE, orientation = NA, se = TRUE, method = function (formula, data, subset, we
## {
##     ret.x <- x
##     ret.y <- y
##     cl <- match.call()
##     mf <- match.call(expand.dots = FALSE)
##     m <- match(c("formula", "data", "subset", "weights", "na.action", "offset"), names(mf), 0)
##     mf <- mf[c(1, m)]
##     mf$drop.unused.levels <- TRUE
```

```
##     mf[[1]] <- quote(stats::model.frame)
##     mf <- eval(mf, parent.frame())
##     if (method == "model.frame")
##         return(mf)
##     else if (method != "qr")
##         warning(gettextf("method = '%s' is not supported. Using 'qr'", method), domain = NA)
##     mt <- attr(mf, "terms")
##     y <- model.response(mf, "numeric")
##     w <- as.vector(model.weights(mf))
##     if (!is.null(w) && !is.numeric(w))
##         stop("'weights' must be a numeric vector")
##     offset <- model.offset(mf)
##     mlm <- is.matrix(y)
##     ny <- if (mlm)
##         nrow(y)
##     else length(y)
##     if (!is.null(offset)) {
##         if (!mlm)
##             offset <- as.vector(offset)
##         if (NROW(offset) != ny)
##             stop(gettextf("number of offsets is %d, should equal %d (number of observations)", NROW(o
##     }
##     if (is.empty.model(mt)) {
##         x <- NULL
##         z <- list(coefficients = if (mlm) matrix(NA, 0, ncol(y)) else numeric(), residuals = y, fitte
##         if (!is.null(offset)) {
##             z$fitted.values <- offset
##             z$residuals <- y - offset
##         }
##     }
##     else {
##         x <- model.matrix(mt, mf, contrasts)
##         z <- if (is.null(w))
##             lm.fit(x, y, offset = offset, singular.ok = singular.ok, ...)
##         else lm.wfit(x, y, w, offset = offset, singular.ok = singular.ok, ...)
##     }
##     class(z) <- c(if (mlm) "mlm", "lm")
##     z$na.action <- attr(mf, "na.action")
##     z$offset <- offset
##     z$contrasts <- attr(x, "contrasts")
##     z$xlevels <- .getXlevels(mt, mf)
##     z$call <- cl
##     z$terms <- mt
##     if (model)
##         z$model <- mf
##     if (ret.x)
##         z$x <- x
##     if (ret.y)
##         z$y <- y
##     if (!qr)
##         z$qr <- NULL
##     z
## }
## position_identity
```

```r
#PM10
  ggplot(combined_dataset, aes(y = PM10, x = first.month, colour = Location)) +
  geom_line() +
  scale_x_date(date_breaks = "10 years", date_labels = "%Y")
```

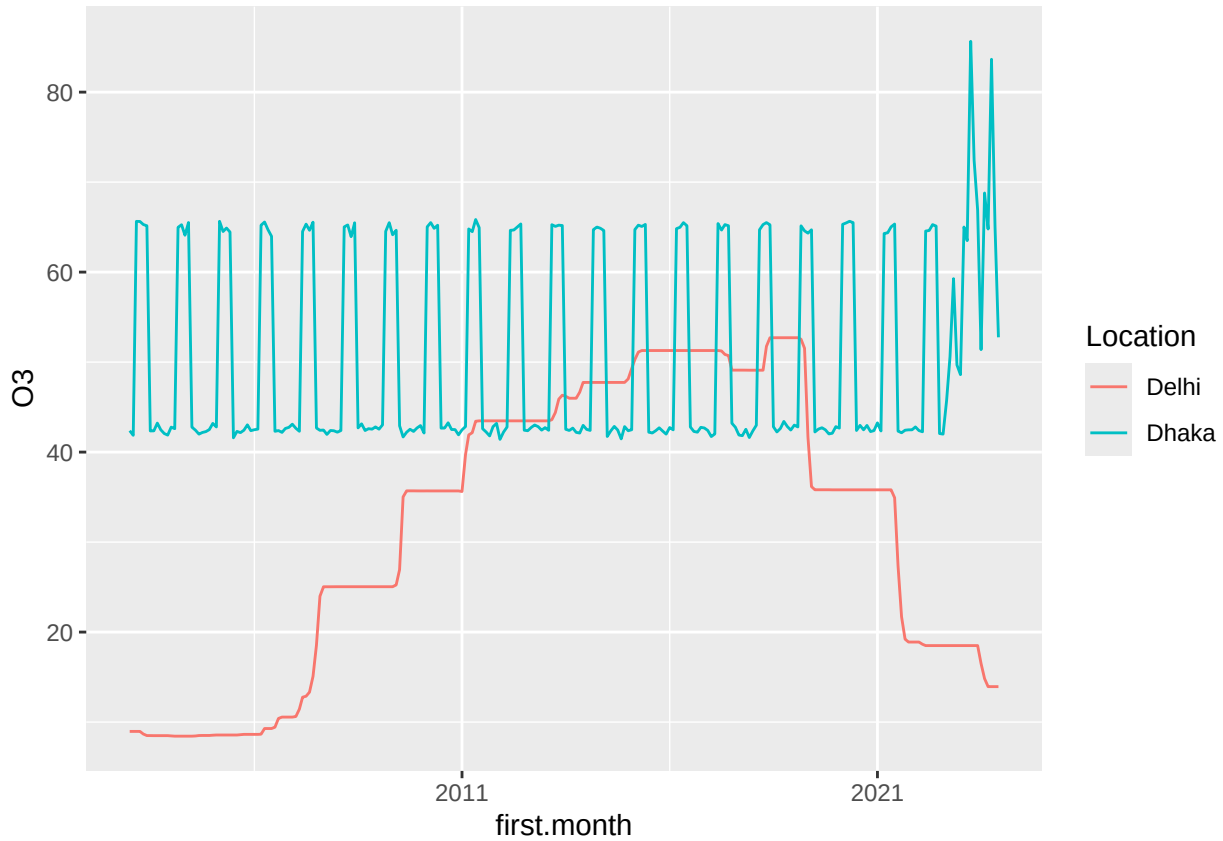



```r
  geom_smooth(method = lm)
```

```
## geom_smooth: na.rm = FALSE, orientation = NA, se = TRUE
## stat_smooth: na.rm = FALSE, orientation = NA, se = TRUE, method = function (formula, data, subset, we
## {
##     ret.x <- x
##     ret.y <- y
##     cl <- match.call()
##     mf <- match.call(expand.dots = FALSE)
##     m <- match(c("formula", "data", "subset", "weights", "na.action", "offset"), names(mf), 0)
##     mf <- mf[c(1, m)]
##     mf$drop.unused.levels <- TRUE
##     mf[[1]] <- quote(stats::model.frame)
##     mf <- eval(mf, parent.frame())
##     if (method == "model.frame")
##         return(mf)
##     else if (method != "qr")
##         warning(gettextf("method = '%s' is not supported. Using 'qr'", method), domain = NA)
##     mt <- attr(mf, "terms")
##     y <- model.response(mf, "numeric")
```

```
##     w <- as.vector(model.weights(mf))
##     if (!is.null(w) && !is.numeric(w))
##         stop("'weights' must be a numeric vector")
##     offset <- model.offset(mf)
##     mlm <- is.matrix(y)
##     ny <- if (mlm)
##         nrow(y)
##     else length(y)
##     if (!is.null(offset)) {
##         if (!mlm)
##             offset <- as.vector(offset)
##         if (NROW(offset) != ny)
##             stop(gettextf("number of offsets is %d, should equal %d (number of observations)", NROW(o
##     }
##     if (is.empty.model(mt)) {
##         x <- NULL
##         z <- list(coefficients = if (mlm) matrix(NA, 0, ncol(y)) else numeric(), residuals = y, fitte
##         if (!is.null(offset)) {
##             z$fitted.values <- offset
##             z$residuals <- y - offset
##         }
##     }
##     else {
##         x <- model.matrix(mt, mf, contrasts)
##         z <- if (is.null(w))
##             lm.fit(x, y, offset = offset, singular.ok = singular.ok, ...)
##         else lm.wfit(x, y, w, offset = offset, singular.ok = singular.ok, ...)
##     }
##     class(z) <- c(if (mlm) "mlm", "lm")
##     z$na.action <- attr(mf, "na.action")
##     z$offset <- offset
##     z$contrasts <- attr(x, "contrasts")
##     z$xlevels <- .getXlevels(mt, mf)
##     z$call <- cl
##     z$terms <- mt
##     if (model)
##         z$model <- mf
##     if (ret.x)
##         z$x <- x
##     if (ret.y)
##         z$y <- y
##     if (!qr)
##         z$qr <- NULL
##     z
## }
## position_identity
```

```
#O3
  ggplot(combined_dataset, aes(y = O3, x = first.month, colour = Location)) +
  geom_line() +
  scale_x_date(date_breaks = "10 years", date_labels = "%Y")
```

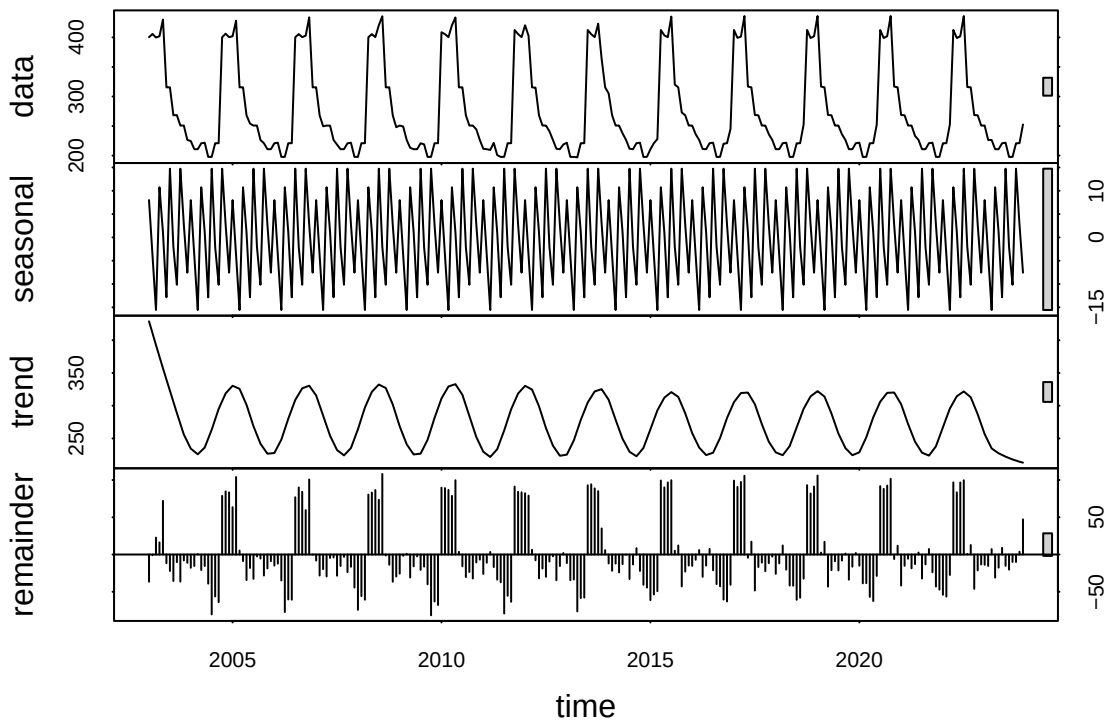



```
  geom_smooth(method = lm)
```

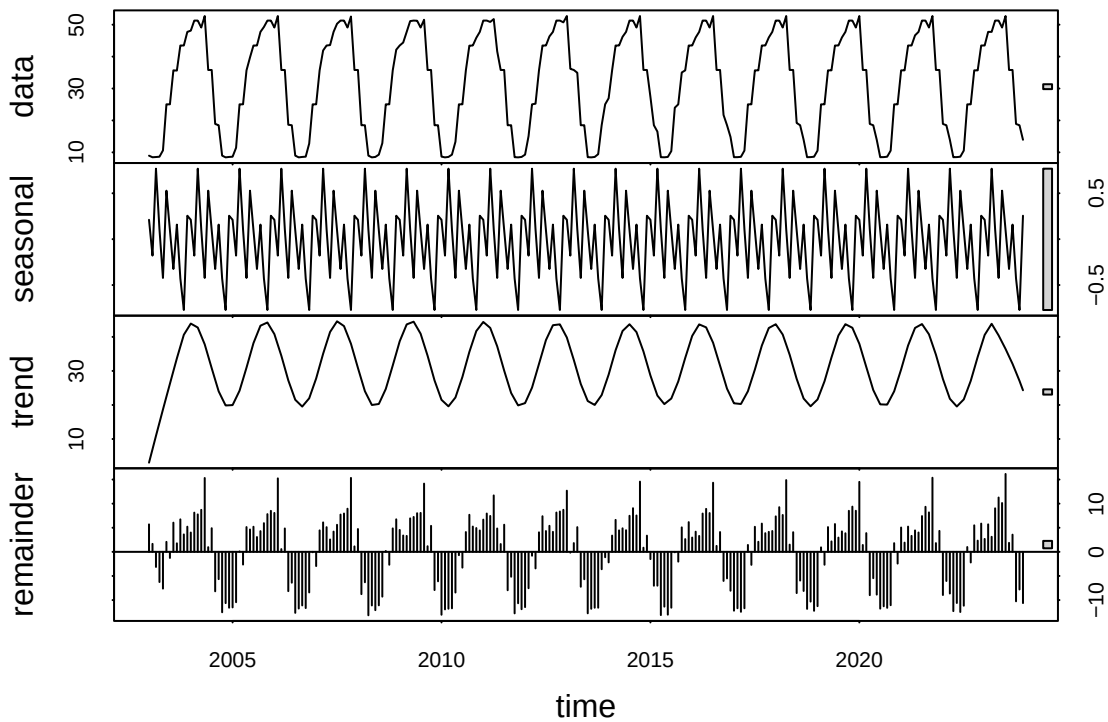
```
## geom_smooth: na.rm = FALSE, orientation = NA, se = TRUE
## stat_smooth: na.rm = FALSE, orientation = NA, se = TRUE, method = function (formula, data, subset, we
## {
##     ret.x <- x
##     ret.y <- y
##     cl <- match.call()
##     mf <- match.call(expand.dots = FALSE)
##     m <- match(c("formula", "data", "subset", "weights", "na.action", "offset"), names(mf), 0)
##     mf <- mf[c(1, m)]
##     mf$drop.unused.levels <- TRUE
##     mf[[1]] <- quote(stats::model.frame)
##     mf <- eval(mf, parent.frame())
##     if (method == "model.frame")
##         return(mf)
##     else if (method != "qr")
##         warning(gettextf("method = '%s' is not supported. Using 'qr'", method), domain = NA)
##     mt <- attr(mf, "terms")
##     y <- model.response(mf, "numeric")
##     w <- as.vector(model.weights(mf))
##     if (!is.null(w) && !is.numeric(w))
##         stop("'weights' must be a numeric vector")
##     offset <- model.offset(mf)
##     mlm <- is.matrix(y)
```

```
##     ny <- if (mlm)
##         nrow(y)
##     else length(y)
##     if (!is.null(offset)) {
##         if (!mlm)
##             offset <- as.vector(offset)
##         if (NROW(offset) != ny)
##             stop(gettextf("number of offsets is %d, should equal %d (number of observations)", NROW(o
##     }
##     if (is.empty.model(mt)) {
##         x <- NULL
##         z <- list(coefficients = if (mlm) matrix(NA, 0, ncol(y)) else numeric(), residuals = y, fitte
##         if (!is.null(offset)) {
##             z$fitted.values <- offset
##             z$residuals <- y - offset
##         }
##     }
##     else {
##         x <- model.matrix(mt, mf, contrasts)
##         z <- if (is.null(w))
##             lm.fit(x, y, offset = offset, singular.ok = singular.ok, ...)
##         else lm.wfit(x, y, w, offset = offset, singular.ok = singular.ok, ...)
##     }
##     class(z) <- c(if (mlm) "mlm", "lm")
##     z$na.action <- attr(mf, "na.action")
##     z$offset <- offset
##     z$contrasts <- attr(x, "contrasts")
##     z$xlevels <- .getXlevels(mt, mf)
##     z$call <- cl
##     z$terms <- mt
##     if (model)
##         z$model <- mf
##     if (ret.x)
##         z$x <- x
##     if (ret.y)
##         z$y <- y
##     if (!qr)
##         z$qr <- NULL
##     z
## }
## position_identity
```

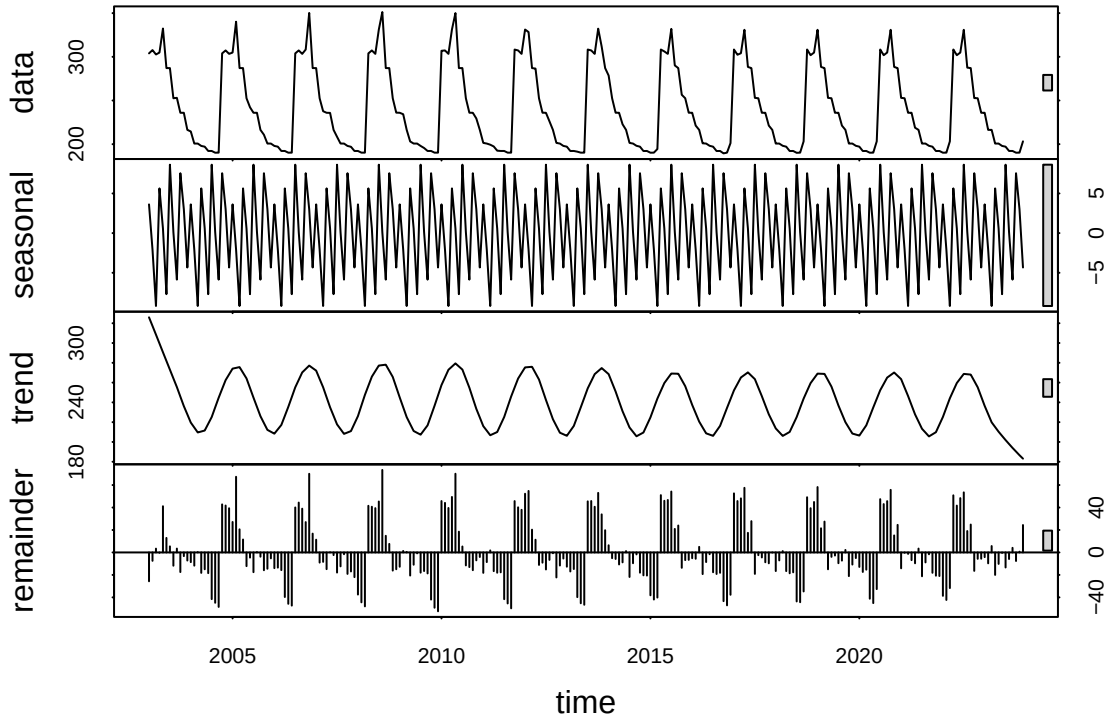
```r
# Monthly AQI for Delhi
Delhi.AQI.monthly.ts <- ts(delhi_final_processed$AQI,
                           start = c(2003,1), frequency = 12)
Delhi.AQI.monthly.decomposed <- stl(Delhi.AQI.monthly.ts,
                                    s.window = "periodic")
plot(Delhi.AQI.monthly.decomposed)
```



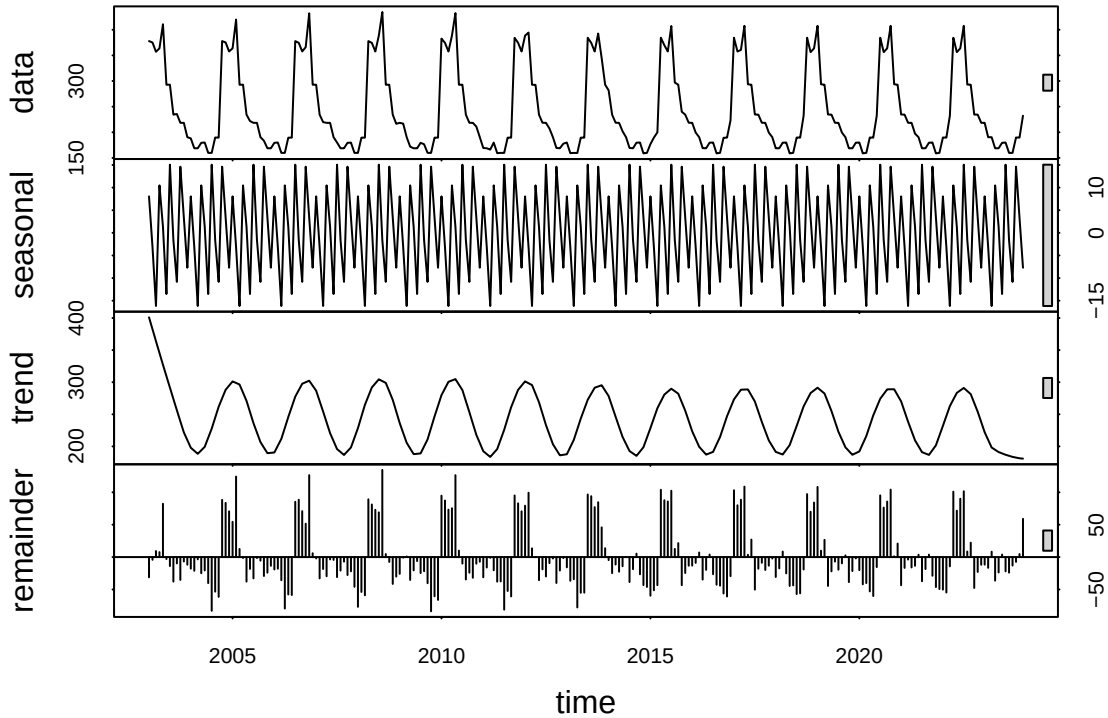

```
# Monthly Ozone for Delhi
Delhi.Ozone.monthly.ts <- ts(delhi_final_processed$O3,
                             start = c(2003,1), frequency = 12)
Delhi.Ozone.monthly.decomposed <- stl(Delhi.Ozone.monthly.ts,
                                      s.window = "periodic")
plot(Delhi.Ozone.monthly.decomposed)
```



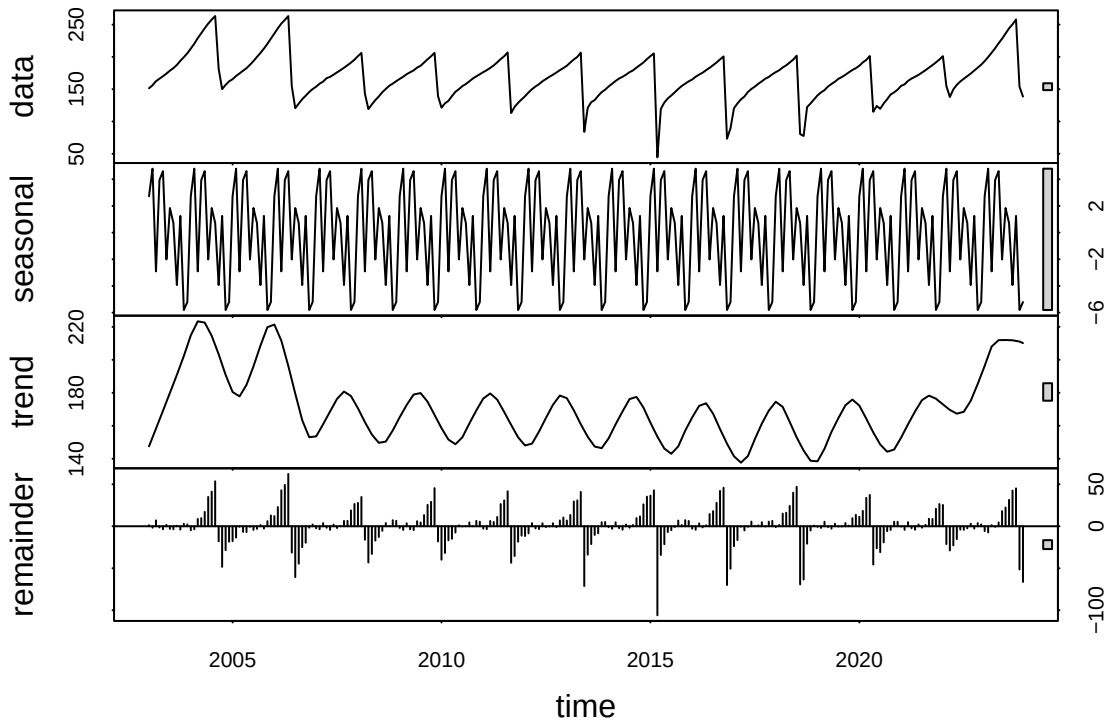

```r
# Monthly PM2.5 for Delhi
Delhi.PM2.5.monthly.ts <- ts(delhi_final_processed$PM2.5,
                             start = c(2003,1), frequency = 12)
Delhi.PM2.5.monthly.decomposed <- stl(Delhi.PM2.5.monthly.ts,
                                      s.window = "periodic")
plot(Delhi.PM2.5.monthly.decomposed)
```



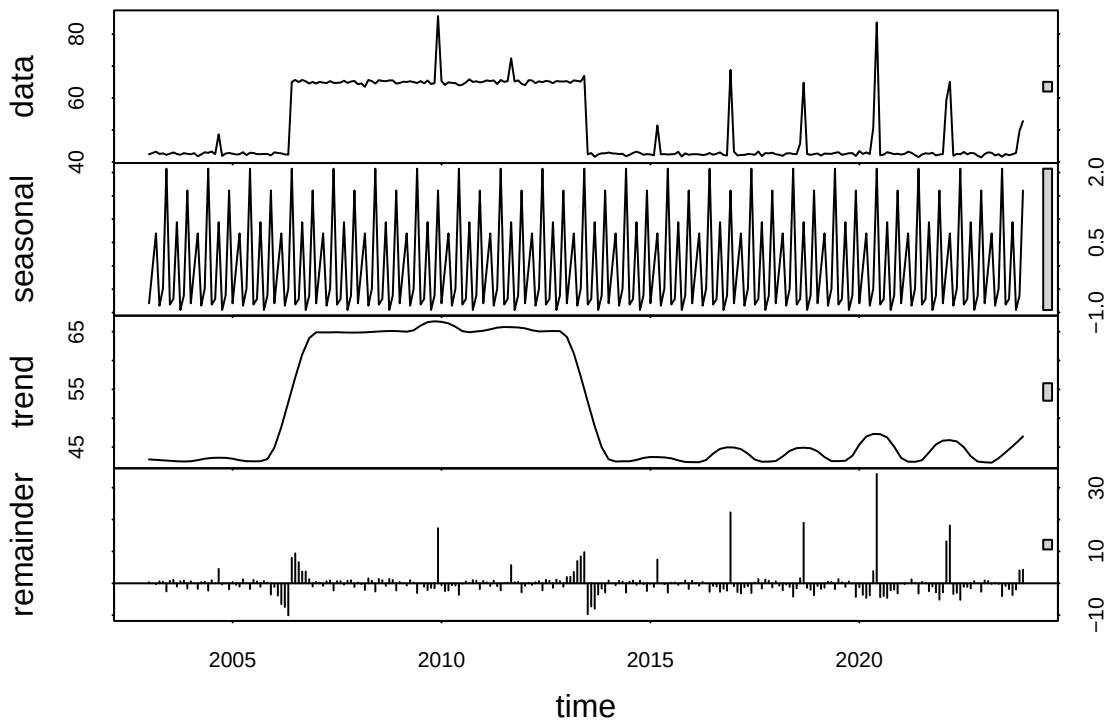

```r
# Monthly PM10 for Delhi
Delhi.PM10.monthly.ts <- ts(delhi_final_processed$PM10,
                            start = c(2003,1), frequency = 12)
Delhi.PM10.monthly.decomposed <- stl(Delhi.PM10.monthly.ts,
                                     s.window = "periodic")
plot(Delhi.PM10.monthly.decomposed)
```



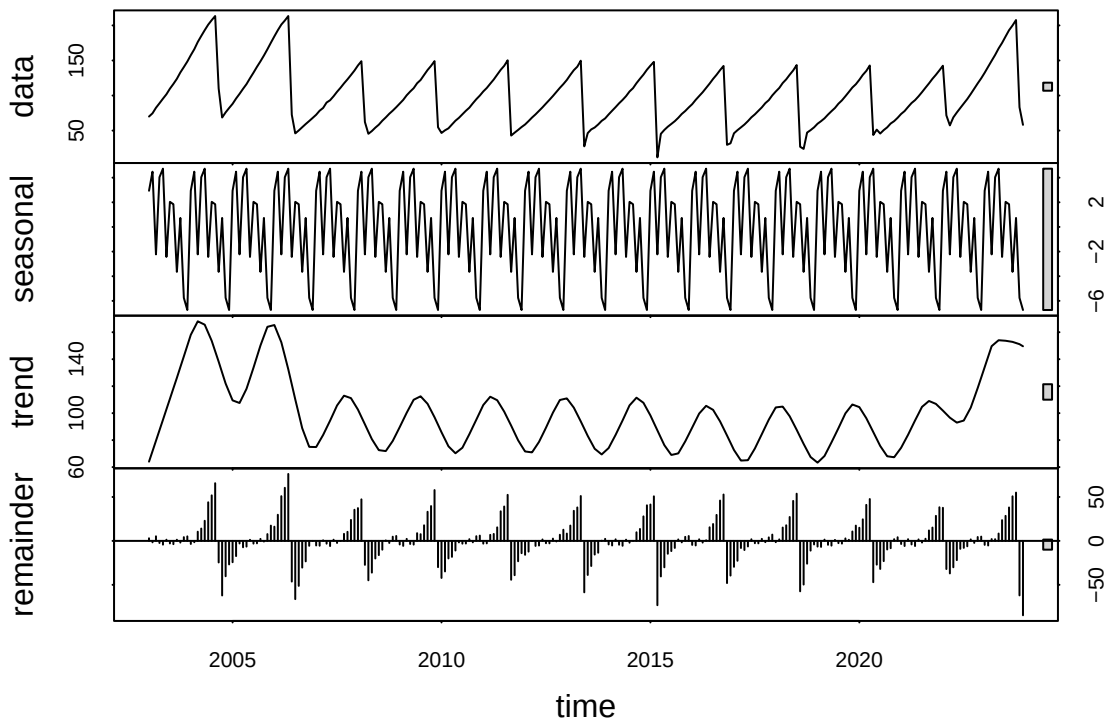

```
# Monthly AQI for Dhaka
Dhaka.AQI.monthly.ts <- ts(dhaka_final_processed$AQI,
                           start = c(2003,1), frequency = 12)
Dhaka.AQI.monthly.decomposed <- stl(Dhaka.AQI.monthly.ts,
                                    s.window = "periodic")
plot(Dhaka.AQI.monthly.decomposed)
```



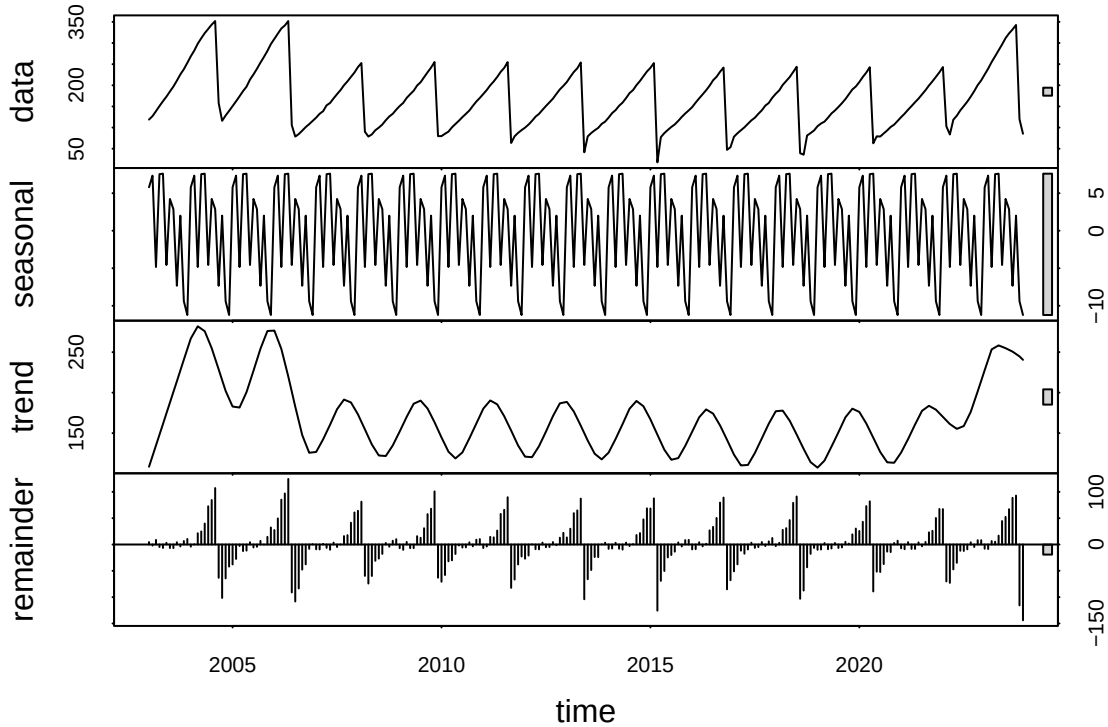

```r
# Monthly Ozone for Dhaka
Dhaka.Ozone.monthly.ts <- ts(dhaka_final_processed$O3,
                             start = c(2003,1), frequency = 12)
Dhaka.Ozone.monthly.decomposed <- stl(Dhaka.Ozone.monthly.ts,
                                      s.window = "periodic")
plot(Dhaka.Ozone.monthly.decomposed)
```




```r
# Monthly PM2.5 for Dhaka
Dhaka.PM2.5.monthly.ts <- ts(dhaka_final_processed$PM2.5,
                             start = c(2003,1), frequency = 12)
Dhaka.PM2.5.monthly.decomposed <- stl(Dhaka.PM2.5.monthly.ts,
                                      s.window = "periodic")
plot(Dhaka.PM2.5.monthly.decomposed)
```




```r
# Monthly PM10 for Dhaka
Dhaka.PM10.monthly.ts <- ts(dhaka_final_processed$PM10,
                            start = c(2003,1), frequency = 12)
Dhaka.PM10.monthly.decomposed <- stl(Dhaka.PM10.monthly.ts,
                                     s.window = "periodic")
plot(Dhaka.PM10.monthly.decomposed)
```




```r
# Seasonal Mann-Kendall Dhaka

# Dhaka AQI
Dhaka.AQI.monthly.trend <- trend::smk.test(Dhaka.AQI.monthly.ts)
Dhaka.AQI.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.AQI.monthly.ts
## z = -1.7696, p-value = 0.0768
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -204 13160
```

```r
summary(Dhaka.AQI.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.AQI.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
```

```
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -2 1096.7 -0.010 -0.030  0.97591
## Season 2:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 3:   S = 0  -14 1096.7 -0.067 -0.393  0.69464
## Season 4:   S = 0  -10 1096.7 -0.048 -0.272  0.78580
## Season 5:   S = 0  -34 1096.7 -0.162 -0.996  0.31901
## Season 6:   S = 0  -20 1096.7 -0.095 -0.574  0.56614
## Season 7:   S = 0   -6 1096.7 -0.029 -0.151  0.87999
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 10:   S = 0   6 1096.7  0.029  0.151  0.87999
## Season 11:   S = 0 -34 1096.7 -0.162 -0.996  0.31901
## Season 12:   S = 0 -30 1096.7 -0.143 -0.876  0.38119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka AQI not significant (p = 0.0768)


# Dhaka PM2.5
Dhaka.PM2.5.monthly.trend <- trend::smk.test(Dhaka.PM2.5.monthly.ts)
Dhaka.PM2.5.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.PM2.5.monthly.ts
## z = -1.7696, p-value = 0.0768
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -204 13160
```

```r
summary(Dhaka.PM2.5.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.PM2.5.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0    0 1096.7  0.000  0.000  1.00000
## Season 2:   S = 0  -28 1096.7 -0.133 -0.815  0.41489
## Season 3:   S = 0  -14 1096.7 -0.067 -0.393  0.69464
## Season 4:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 5:   S = 0  -38 1096.7 -0.181 -1.117  0.26387
## Season 6:   S = 0  -22 1096.7 -0.105 -0.634  0.52599
```

```
## Season 7:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 10:   S = 0   8 1096.7  0.038  0.211  0.83259
## Season 11:   S = 0 -28 1096.7 -0.133 -0.815  0.41489
## Season 12:   S = 0 -24 1096.7 -0.114 -0.695  0.48735
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka PM2.5 not significant (p = 0.0768)

# Dhaka PM10
Dhaka.PM10.monthly.trend <- trend::smk.test(Dhaka.PM10.monthly.ts)
Dhaka.PM10.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.PM10.monthly.ts
## z = -1.6127, p-value = 0.1068
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -186 13160
```

```r
summary(Dhaka.PM10.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.PM10.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0    4 1096.7  0.019  0.091  0.92782
## Season 2:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 3:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 4:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 5:   S = 0  -32 1096.7 -0.152 -0.936  0.34922
## Season 6:   S = 0  -16 1096.7 -0.076 -0.453  0.65058
## Season 7:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0   -2 1096.7 -0.010 -0.030  0.97591
## Season 10:   S = 0   6 1096.7  0.029  0.151  0.87999
## Season 11:   S = 0 -34 1096.7 -0.162 -0.996  0.31901
## Season 12:   S = 0 -28 1096.7 -0.133 -0.815  0.41489
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka PM2.5 not significant (p = 0.1068)

# Dhaka O3
Dhaka.Ozone.monthly.trend <- trend::smk.test(Dhaka.Ozone.monthly.ts)
Dhaka.Ozone.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.Ozone.monthly.ts
## z = -4.8206, p-value = 1.432e-06
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -554 13160
```

```r
summary(Dhaka.Ozone.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.Ozone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0  -30 1096.7 -0.143 -0.876 0.381187
## Season 2:   S = 0  -38 1096.7 -0.181 -1.117 0.263872
## Season 3:   S = 0    6 1096.7  0.029  0.151 0.879988
## Season 4:   S = 0  -56 1096.7 -0.267 -1.661 0.096747 .
## Season 5:   S = 0  -54 1096.7 -0.257 -1.600 0.109502
## Season 6:   S = 0  -40 1096.7 -0.190 -1.178 0.238924
## Season 7:   S = 0  -70 1096.7 -0.333 -2.084 0.037198 *
## Season 8:   S = 0  -76 1096.7 -0.362 -2.265 0.023527 *
## Season 9:   S = 0  -62 1096.7 -0.295 -1.842 0.065473 .
## Season 10:   S = 0 -12 1096.7 -0.057 -0.332 0.739764
## Season 11:   S = 0 -66 1096.7 -0.314 -1.963 0.049669 *
## Season 12:   S = 0 -56 1096.7 -0.267 -1.661 0.096747 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka ozone (O3) is significant (1.432e-06)

# Seasonal Mann-Kendall Delhi

# Delhi AQI
Delhi.AQI.monthly.trend <- trend::smk.test(Delhi.AQI.monthly.ts)
Delhi.AQI.monthly.trend
```

```
##
```

```
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.AQI.monthly.ts
## z = -0.98905, p-value = 0.3226
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##  -114.00 13053.33
```

```r
summary(Delhi.AQI.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.AQI.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -8 1092.0 -0.038 -0.212  0.83224
## Season 2:   S = 0  -41 1088.3 -0.199 -1.212  0.22532
## Season 3:   S = 0    1 1085.7  0.005  0.000  1.00000
## Season 4:   S = 0   -6 1092.0 -0.029 -0.151  0.87973
## Season 5:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 6:   S = 0    9 1083.7  0.044  0.243  0.80799
## Season 7:   S = 0   16 1090.0  0.077  0.454  0.64959
## Season 8:   S = 0  -27 1088.3 -0.131 -0.788  0.43063
## Season 9:   S = 0    3 1083.7  0.015  0.061  0.95155
## Season 10:   S = 0  -4 1092.0 -0.019 -0.091  0.92766
## Season 11:   S = 0 -36 1084.7 -0.176 -1.063  0.28791
## Season 12:   S = 0  10 1084.7  0.049  0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi AQI not significant (0.3226)

# Delhi PM2.5
Delhi.PM2.5.monthly.trend <- trend::smk.test(Delhi.PM2.5.monthly.ts)
Delhi.PM2.5.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.PM2.5.monthly.ts
## z = -1.7156, p-value = 0.08624
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##  -197.00 13052.33
```

```r
summary(Delhi.PM2.5.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.PM2.5.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0  -24 1092.0 -0.115 -0.696  0.48642
## Season 2:   S = 0  -41 1088.3 -0.199 -1.212  0.22532
## Season 3:   S = 0  -13 1085.7 -0.063 -0.364  0.71571
## Season 4:   S = 0  -14 1092.0 -0.067 -0.393  0.69403
## Season 5:   S = 0  -19 1088.3 -0.092 -0.546  0.58533
## Season 6:   S = 0    3 1083.7  0.015  0.061  0.95155
## Season 7:   S = 0   -4 1090.0 -0.019 -0.091  0.92760
## Season 8:   S = 0  -13 1088.3 -0.063 -0.364  0.71605
## Season 9:   S = 0   -1 1083.7 -0.005  0.000  1.00000
## Season 10:   S = 0 -24 1092.0 -0.115 -0.696  0.48642
## Season 11:   S = 0 -37 1083.7 -0.181 -1.094  0.27413
## Season 12:   S = 0 -10 1084.7 -0.049 -0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi AQI not significant (0.08624)

# Delhi PM10
Delhi.PM10.monthly.trend <- trend::smk.test(Delhi.PM10.monthly.ts)
Delhi.PM10.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.PM10.monthly.ts
## z = -0.79649, p-value = 0.4257
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##   -92.00 13053.33
```

```r
summary(Delhi.PM10.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.PM10.monthly.ts
## alternative hypothesis: two.sided
##
```

```
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -2 1092.0 -0.010 -0.030  0.97586
## Season 2:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 3:   S = 0   -5 1085.7 -0.024 -0.121  0.90338
## Season 4:   S = 0    2 1092.0  0.010  0.030  0.97586
## Season 5:   S = 0  -25 1088.3 -0.121 -0.727  0.46692
## Season 6:   S = 0    7 1083.7  0.034  0.182  0.85537
## Season 7:   S = 0   16 1090.0  0.077  0.454  0.64959
## Season 8:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 9:   S = 0   -5 1083.7 -0.024 -0.122  0.90329
## Season 10:   S = 0   2 1092.0  0.010  0.030  0.97586
## Season 11:   S = 0 -30 1084.7 -0.146 -0.881  0.37857
## Season 12:   S = 0  10 1084.7  0.049  0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi PM10 not significant (0.4257)

# Delhi O3
Delhi.Ozone.monthly.trend <- trend::smk.test(Delhi.Ozone.monthly.ts)
Delhi.Ozone.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.Ozone.monthly.ts
## z = 1.0157, p-value = 0.3098
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##   117.00 13044.33
```

```r
summary(Delhi.Ozone.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.Ozone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   25 1091.0  0.120  0.727  0.46747
## Season 2:   S = 0   11 1088.3  0.053  0.303  0.76180
## Season 3:   S = 0   14 1084.7  0.068  0.395  0.69305
## Season 4:   S = 0   19 1091.0  0.092  0.545  0.58578
## Season 5:   S = 0    9 1088.3  0.044  0.242  0.80839
## Season 6:   S = 0   14 1082.7  0.069  0.395  0.69278
```

```
## Season 7:   S = 0   21 1089.0  0.102  0.606  0.54447
## Season 8:   S = 0    8 1087.3  0.039  0.212  0.83189
## Season 9:   S = 0    9 1083.7  0.044  0.243  0.80799
## Season 10:   S = 0   5 1091.0  0.024  0.121  0.90361
## Season 11:   S = 0  -7 1083.7 -0.034 -0.182  0.85537
## Season 12:   S = 0 -11 1083.7 -0.054 -0.304  0.76130
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi ozone (O3) not significant (0.3098)


# Non-Seasonal Mann-Kendall Dhaka

# Dhaka AQI
Dhaka.monthly.AQI.nonseasonal <- Dhaka.AQI.monthly.ts -
  Dhaka.AQI.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.AQI.trend <- Kendall::MannKendall(
  Dhaka.monthly.AQI.nonseasonal)
Dhaka.nonseasonal.AQI.trend
```

```
## tau = -0.0537, 2-sided pvalue =0.20448
```

```r
summary(Dhaka.nonseasonal.AQI.trend)
```

```
## Score =  -1698 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0537, 2-sided pvalue =0.20448
```

```r
# Dhaka PM2.5
Dhaka.monthly.PM2.5.nonseasonal <- Dhaka.PM2.5.monthly.ts -
  Dhaka.PM2.5.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.PM2.5.trend <- Kendall::MannKendall(
  Dhaka.monthly.PM2.5.nonseasonal)
Dhaka.nonseasonal.PM2.5.trend
```

```
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
summary(Dhaka.nonseasonal.PM2.5.trend)
```

```
## Score =  -1588 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
# Dhaka PM10
Dhaka.monthly.PM10.nonseasonal <- Dhaka.PM10.monthly.ts -
  Dhaka.PM10.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.PM10.trend <- Kendall::MannKendall(
  Dhaka.monthly.PM10.nonseasonal)
Dhaka.nonseasonal.PM10.trend
```

```
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
summary(Dhaka.nonseasonal.PM10.trend)
```

```
## Score =  -1588 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
# Dhaka O3
Dhaka.monthly.Ozone.nonseasonal <- Dhaka.Ozone.monthly.ts -
  Dhaka.Ozone.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.Ozone.trend <- Kendall::MannKendall(
  Dhaka.monthly.Ozone.nonseasonal)
Dhaka.nonseasonal.Ozone.trend
```

```
## tau = -0.201, 2-sided pvalue =2.0616e-06
```

```r
summary(Dhaka.nonseasonal.Ozone.trend)
```

```
## Score =  -6350 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.201, 2-sided pvalue =2.0616e-06
```

```r
# Non-Seasonal Mann-Kendall Delhi

# Delhi AQI
Delhi.monthly.AQI.nonseasonal <- Delhi.AQI.monthly.ts -
  Delhi.AQI.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.AQI.trend <- Kendall::MannKendall(
  Delhi.monthly.AQI.nonseasonal)
Delhi.nonseasonal.AQI.trend
```

```
## tau = -0.0681, 2-sided pvalue =0.10775
```

```r
summary(Delhi.nonseasonal.AQI.trend)
```

```
## Score =  -2152 , Var(Score) = 1788519
## denominator =  31580.97
## tau = -0.0681, 2-sided pvalue =0.10775
```

```r
# Delhi PM2.5
Delhi.monthly.PM2.5.nonseasonal <- Delhi.PM2.5.monthly.ts -
  Delhi.PM2.5.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.PM2.5.trend <- Kendall::MannKendall(
  Delhi.monthly.PM2.5.nonseasonal)
Delhi.nonseasonal.PM2.5.trend
```

```
## tau = -0.0903, 2-sided pvalue =0.03296
```

```r
summary(Delhi.nonseasonal.PM2.5.trend)
```

```
## Score =  -2853 , Var(Score) = 1788518
## denominator =  31580.47
## tau = -0.0903, 2-sided pvalue =0.03296
```

```r
# Delhi PM10
Delhi.monthly.PM10.nonseasonal <- Delhi.PM10.monthly.ts -
  Delhi.PM10.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.PM10.trend <- Kendall::MannKendall(
  Delhi.monthly.PM10.nonseasonal)
Delhi.nonseasonal.PM10.trend
```

```
## tau = -0.0624, 2-sided pvalue =0.14094
```

```r
summary(Delhi.nonseasonal.PM10.trend)
```

```
## Score =  -1970 , Var(Score) = 1788519
## denominator =  31580.97
## tau = -0.0624, 2-sided pvalue =0.14094
```

```r
# Delhi O3
Delhi.monthly.Ozone.nonseasonal <- Delhi.Ozone.monthly.ts -
  Delhi.Ozone.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.Ozone.trend <- Kendall::MannKendall(
  Delhi.monthly.Ozone.nonseasonal)
Delhi.nonseasonal.Ozone.trend
```

```
## tau = 0.0326, 2-sided pvalue =0.44208
```

```r
summary(Delhi.nonseasonal.Ozone.trend)
```

```
## Score =  1029 , Var(Score) = 1788510
## denominator =  31576.46
## tau = 0.0326, 2-sided pvalue =0.44208
```

Table 1: Summary of Mann-Kendall Results

Location	Variable	Seasonal p-value	Non-Seasonal p-value
Dhaka	AQI	0.0768	0.2045
Dhaka	PM 2.5	0.0768	0.2354
Dhaka	PM 10	0.1068	0.2354
Dhaka	O3	**1.43e-06**	**2.06e-06**
Delhi	AQI	0.3226	0.1078
Delhi	PM 2.5	0.0862	**0.0330**
Delhi	PM 10	0.4257	0.1409

Location	Variable	Seasonal p-value	Non-Seasonal p-value
Delhi	O3	0.3098	0.4421

```r
# AQI
t.test(AQI ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  AQI by Location
## t = 19.658, df = 344.72, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##   94.82248 115.90689
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            275.7138            170.3491
```

```r
# PM2.5
t.test(PM2.5 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  PM2.5 by Location
## t = 35.011, df = 490.03, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  132.4543 148.2050
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            240.5413            100.2117
```

```r
# PM10
t.test(PM10 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  PM10 by Location
## t = 10.773, df = 490.34, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  59.67191 86.29319
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            242.1669            169.1844
```

```r
# O3
t.test(O3 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  O3 by Location
## t = -15.654, df = 450.94, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  -21.48934 -16.69565
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            31.78149            50.87398
```

```r
#Linear Regression

# AQI
AQI.model <- lm(AQI ~ first.month * Location, data = combined_dataset)
summary(AQI.model)
```

```
##
## Call:
## lm(formula = AQI ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -152.894  -19.418   -1.296   16.357   93.331
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.595e+02  1.582e+01   48.00   <2e-16 ***
## first.month               -3.048e-02  9.873e-04  -30.87   <2e-16 ***
## LocationDhaka             -7.062e+02  2.238e+01  -31.56   <2e-16 ***
## first.month:LocationDhaka  3.785e-02  1.396e-03   27.11   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.7 on 500 degrees of freedom
## Multiple R-squared:  0.8128, Adjusted R-squared:  0.8117
## F-statistic: 723.5 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# PM2.5
PM2.5.model <- lm(PM2.5 ~ first.month * Location, data = combined_dataset)
summary(PM2.5.model)
```

```
##
## Call:
## lm(formula = PM2.5 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -127.439  -13.193   -3.511    9.854   79.986
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)                5.623e+02  1.238e+01   45.42   <2e-16 ***
## first.month               -2.027e-02  7.725e-04  -26.24   <2e-16 ***
## LocationDhaka             -6.310e+02  1.751e+01  -36.04   <2e-16 ***
## first.month:LocationDhaka  3.091e-02  1.092e-03   28.30   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.15 on 500 degrees of freedom
## Multiple R-squared:  0.8946, Adjusted R-squared:  0.894
## F-statistic:  1415 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# PM10
PM10.model <- lm(PM10 ~ first.month * Location, data = combined_dataset)
summary(PM10.model)
```

```
##
## Call:
## lm(formula = PM10 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -214.902  -26.310   -5.708   23.492  128.747
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.438e+02  2.324e+01   32.00   <2e-16 ***
## first.month               -3.160e-02  1.450e-03  -21.79   <2e-16 ***
## LocationDhaka             -8.491e+02  3.287e+01  -25.83   <2e-16 ***
## first.month:LocationDhaka  4.890e-02  2.051e-03   23.84   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50.98 on 500 degrees of freedom
## Multiple R-squared:  0.6364, Adjusted R-squared:  0.6342
## F-statistic: 291.7 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# O3
O3.model <- lm(O3 ~ first.month * Location, data = combined_dataset)
summary(O3.model)
```

```
##
## Call:
## lm(formula = O3 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.486  -9.189  -4.640  13.627  32.740
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -2.075e+01  5.773e+00  -3.595 0.000356 ***
## first.month                3.310e-03  3.602e-04   9.189  < 2e-16 ***
## LocationDhaka              6.263e+01  8.164e+00   7.671 8.97e-14 ***
```

```
## first.month:LocationDhaka -2.743e-03  5.094e-04  -5.384 1.12e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.66 on 500 degrees of freedom
## Multiple R-squared:  0.4275, Adjusted R-squared:  0.4241
## F-statistic: 124.5 on 3 and 500 DF,  p-value: < 2.2e-16
```